

SHASWAT GLASS SOLUTION PRIVATE LIMITED

TECHNICAL DESIGN MANUAL

HIGH-PERFORMANCE STRUCTURAL & SAFETY GLASS SOLUTIONS



SHASWAT GLASS SOLUTION PRIVATE LIMITED

Established 2022 | Quality Certified: IS 2553 : Part 1 : 2018

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MANUFACTURER NAME: SHASWAT GLASS SOLUTION PVT LTD		DOCUMENT TYPE: TECHNICAL SPECIFICATION SHEET		DOCUMENT ID: SG-TDD-2026-V1.0	
1.0 COMPANY PROFILE & INFRASTRUCTURE INDEX					
Shaswat Glass Solution Private Limited is a premier glass processing manufacturer established to serve the growing demands of the architectural facade, structural safety, and high-end interior industries. Our manufacturing facility is equipped with automated European machines, enabling us to deliver absolute precision, dimensional accuracy, and thermal strength. We are dedicated to safety compliance and are proud to be certified under **IS 2553: Part 1:2018**, which dictates the standards for safety glazing materials in land transit and structural configurations. Our Convection Tempering furnace ensures optimal flatness and minimal optical distortion, making our glass the standard choice for high-rise commercial structures and contemporary residential spaces.			Key Capabilities - CNC Glass Cutting Line - Diamond Double Edging - Convection Tempering Furnace - Insulating DGU Processing - Laminated Safety Line Technical Index Pg 3: Toughened Glass Specs Pg 4: Insulated DGU Specs Pg 5: Low-E Glass Specs Pg 6: Specialty Glass Ranges Pg 7: Quality Testing Parameters Pg 8-9: Machine Facilities		
ORGANIZATIONAL STATEMENT & COMMITMENT:					
Shaswat Glass aims to innovate structural safety glazing. We continually adapt our convective heating parameters and tooling tolerances to exceed standard specifications and deliver elite materials for contemporary facades.					
SHEET SUBJECT: PRODUCT: COMPANY PROFILE & INDEX		STANDARDS & CERTIFICATIONS: IS 2553-1 EN 12150-1 ASTM C1048		DOCUMENT TRACKING: REV: 00 SHEET 2 OF 10	

2.0 TOUGHENED / TEMPERED SAFETY GLASS SPECIFICATIONS



Shaswat Toughened Glass is a high-strength safety glass processed by controlled thermal treatments. By heating flat annealed glass to approximately 620°C and then rapidly cooling it with high-pressure air, the glass gains a highly compressed outer layer. This compression gives the glass its superior load-bearing capacity—making it up to 5 times stronger than standard annealed glass. If broken, it shatters into small, blunt-edged fragments, minimizing severe cuts.

Applications:

- Facades, glass doors & partitions
- Balustrades, railings, canopies & skylights
- Bathroom glass fixtures & shower doors

Technical Specifications:

Thickness Range	3.5 mm to 19 mm
Max Size	2440 mm × 5000 mm
Min Size	100 mm × 300 mm
Standards	IS 2553: Part 1 : 2018 Certified

ENGINEERING DESIGN ADVISORY:

Toughened glass panels cannot be cut, drilled, or edge-polished after the thermal tempering cycle. All sheet sizes, notches, countersinks, and edge grinding profiles must be fully fabricated and validated prior to furnace entry.

3.0 DOUBLE GLAZED INSULATING GLASS (DGU) SPECIFICATIONS



Double Glazed Units (DGU) are thermal insulated structures composed of two sheets of glass, separated by an aluminum spacer bar filled with desiccant and sealed hermetically. The air gap creates a thermal barrier, reducing heat gain in summer and loss in winter, while dampening ambient noise.

Technical Specifications:

Pane Thickness	4 mm to 12 mm per pane
Spacer Width	6mm, 9mm, 12mm, 16mm, 20mm
Sealants	Butyl PIB & Polyurethane/Silicone
Acoustic Rating	Reduces external noise by up to 35 dB

Applications:

- Commercial facade curtain walls
- Soundproof windows for hotels & offices
- Industrial cold storage windows

THERMAL DEFLECTION WARNING:

For insulated units deployed in high-altitude zones, pressure equalization tubes (capillaries) must be installed inside the spacer bar frame. This equalizes atmospheric pressure, preventing glass bowing, optical distortion, or edge seal failures.

4.0 LOW-EMISSIVITY (LOW-E) COATING PERFORMANCE



Low-E Coating Glass features a microscopic metallic oxide coating that reflects thermal infrared energy while letting visible light pass. This controls solar heat gain, reducing HVAC cooling bills during summers and retaining interior heat during winters, while protecting indoor furniture from UV fading.

Technical Specifications:

Coating Process	Online Hard Coat Offline Soft Coat
Glass Thickness	5 mm to 12 mm
Performance	Blocks up to 85% Infrared, 80% UV
U-Value Index	Up to 70% better than annealed glass

Applications:

- Double glazed windows in LEED structures
- Commercial facade structural glass panels
- Climate-controlled server rooms & entries

COATING POSITION & HANDLING SPECIFICATION:

To protect sputtered Low-E thin films from physical wear, the coating must face the sealed internal gap of the DGU (Surface #2 or #3). Avoid exposure to acid washes, abrasive scrapers, or prolonged humidity during site storage before installation.

5.0 SPECIALTY GLASS: HEAT-RESISTANT CERAMIC & SKYLIGHT



Ceramic Glass (Heat Resistant)

Ceramic safety glass handles high operating temperatures (up to 700°C) and sudden thermal shocks. **Thickness:** 4mm – 8mm **Applications:** Kiln windows, ovens, furnace partitions, and fireplace screens.



Skylight Laminated Glass

Laminated glass with a tough PVB interlayer. If broken, glass fragments stay adhered to prevent falling hazards. **Thickness:** 6.38mm to 19mm custom laminates **Applications:** Overhead glazing, mall atrium roofs, skylights, and glass walkways.

OVERHEAD SAFETY CODE REGULATION:

National building codes mandate that overhead skylight glazing must use laminated glass containing a PVB interlayer. This prevents tempered glass fragments from falling down as a cluster if struck by hard debris.

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6.0 QUALITY POLICY & NATIONAL SAFETY STANDARDS COMPLIANCE			
Quality Control System: Raw glass undergoes thickness checks before cutting. Edge polishing eliminates edge micro-fractures, and the convection tempering furnace uses automatic temperature controls. We conform strictly to the national standard **IS 2553: Part 1 : 2018** to guarantee safety under mechanical impact. Certifications & Approvals: • Bureau of Indian Standards (BIS) IS 2553: Part 1 certified. • Tempering thermal heat cycle certified. • Standard size and dimensional tolerance control.		Testing Parameter	Quality Assurance Standards
		Impact Strength	Steel ball drop test to evaluate structural shock resistance.
		Fragmentation	Counts broken pieces in 50x50mm area; requires 40+ safety granules.
		Optical Test	Inline reflective scan to guarantee distortion-free facade clarity.
		Thermal Shock	Thermal endurance validated by rapid heating and water quenching.
QUALITY CONTROL AUDIT SHIFT FREQUENCY:			
Destructive fragmentation and optical distortion checks are executed on trial samples at the start of every production shift for each glass thickness profile to guarantee full compliance with safety benchmarks.			
SHEET SUBJECT: PRODUCT: QUALITY ASSURANCE STANDARDS & TESTING		STANDARDS & CERTIFICATIONS: IS 2553-1 EN 12150-1 ASTM C1048	DOCUMENT TRACKING: REV: 00 SHEET 7 OF 10

7.0 INFRASTRUCTURE: CNC PRECISION CUTTING FACILITIES



Intermac CNC Glass Cutting System Shaswat Glass utilizes the industry-leading **Intermac CNC Cutting Machine**, a computer-controlled cutting table. The automated cutting table imports CAD designs directly, processing shapes with extreme speed and zero-defect consistency. Proper cutting is necessary before tempering, ensuring raw glass edges are free of micro-fractures which prevents breakage in the furnace.

Machine Capabilities:

Precision Tolerance	+/- 0.2 mm
Maximum Sheet Size	2440 mm × 5000 mm
Cutting Shapes	Polygons, circles, custom curves
CAD Automation	Direct file interface feed

Equipment Advantages:

- Eliminates manual glass cutting errors
- Maximizes sheet yield and minimizes waste
- Ideal for large-scale custom facade panels

DIAMOND CUTTING TOOL MAINTENANCE:

Micro-fracturing during mechanical cutting creates high stress points which cause panel breakages inside the furnace. CNC diamond scribe wheels are checked and replaced regularly to guarantee perfectly smooth glass edges.

8.0 INFRASTRUCTURE: DIAMOND EDGE POLISHING FACILITIES



SK Glass Edge Polishing Line After cutting, glass edges have micro-flaws. We utilize the automated **SK Glass Edge Polishing Machine** to grind and polish glass edges. This double-edging machine processes opposite sides of glass panels simultaneously, using multi-stage diamond abrasive wheels to grind edges, followed by polishing pads to achieve a clean margin. This prevents breakage in the convection tempering furnace.

Machine Capabilities:

Thickness Range	3.5 mm to 19 mm
Edge Profiles	Flat polished, pencil profiles, bevels
Finish Quality	High gloss mirror-edge finish
Speed Automation	Inline continuous speed feed

Equipment Advantages:

- Eliminates risk of micro-cracks along the edge
- Provides a visually stunning gloss margin for frameless glass
- High production rate for bulk architectural orders

EDGE GRINDING SPECIFICATION FOR TEMPERING:

Tempering standards require flat polished edges with arris (pencil/chamfer margins) to remove razor-sharp edge stress. Never subject raw ground glass to tempering without beveling edge arris margins.

SHASWAT GLASS SOLUTION PRIVATE LIMITED

Redefining Architectural Spaces with Strength & Clarity

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WARNING: SAFETY GLAZING MATERIALS MUST BE INSTALLED ACCORDING TO IS 2553 CODE.